TS3000 series Robot Controller

INSTRUCTION MANUAL

Sensor-less Compliance Control

Rev.1
April 1, 2020

Notice

- Make sure that this instruction manual is delivered to the final user of SHIBAURA MACHINE's industrial robot.
- Before operating the industrial robot, read through and completely understand this manual.

Shibaura Machine

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The information contained in this manual is subject to change without prior notice to effect improvements.

Cautions on Safety

This manual contains the important information on the robot and controller to prevent injury to the operators and persons nearby, to prevent damages to assets and to assure correct use.
Make sure that you fully understand the following details (indications and symbols) before reading this manual. Always observe the information that is noted.

[Explanation of indications]

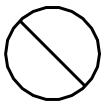
Indication	Meaning of indication
 **DANGER**	This means that "incorrect handling will lead to fatalities or serious injuries."
 **CAUTION**	This means that "incorrect handling may lead to personal injuries *1) or physical damage *2)."

*1) Injuries refer to injuries, burns and electric shocks, etc., which do not require hospitalization or long-term medical treatment.
*2) Physical damage refers to damages due to destruction of assets or resources.

[Explanation of symbols]

Symbol	Meaning of symbol
	This means that the action is prohibited (must not be done). Details of the actions actually prohibited are indicated with pictures or words in or near the symbol.
	This means that the action is mandatory (must be done). Details of the actions that must be done are indicated with pictures or words in or near the symbol.
	This means danger. Details of the actual danger are indicated with pictures or words in or near the symbol.
	This means caution. Details of the actual caution are indicated with pictures or words in or near the symbol.

[Regarding operation]

 **DANGER**	
 Prohibited	• During operation, NEVER enter the dangerous area of the robot. Otherwise, you will be injured seriously. • DO NOT leave in the working range any machinery or materials which will hinder the operation. If a problem occurs in the equipment, a person nearby will be injured or get involved in an accident. • Anyone other than the operator MUST NOT approach the equipment. Should he negligently touch the dangerous part of the equipment, he will get injured or be involved in a serious accident. • NEVER perform an inappropriate operation which is not described in the instruction manual. Otherwise, the equipment will start by mistake, resulting in personal injury or serious accident.
 Danger	• If you feel even a little that you are exposed to danger or that the equipment works abnormally, press the EMERGENCY stop pushbutton switch to stop the equipment. If the equipment is used as it is, you will be injured or involved in a serious accident. When this happens, ask our after-sale service agent for repair. • During operation, be sure to close the equipment cover. Should the cover be opened during operation, you will be struck by an electric shock or get injured. • Only a well-trained and qualified person is allowed to perform the operation. Should the equipment be operated improperly, it will start by mistake, causing a personal injury or serious accident. • If the equipment has malfunctioned, turn the power off, identify and remove the cause of the abnormality, maintain the peripheral equipment and completely restore the malfunctioned equipment. Then start the equipment at slow speed. If the equipment starts, with the abnormality remaining, you may be involved in a serious accident.

CAUTION	
Prohibited	• DO NOT change the data of the system structure file. Otherwise, the robot will move abnormally, resulting in damage or an accident.
Mandatory	• In principle, teaching operation should be performed outside the dangerous area of the robot. If it should be performed inevitably within the dangerous area, strictly observe the following matters. (1) The teaching operation should always be performed by 2 persons. One person performs the job and the other person watches outside the dangerous area. Also, both persons should try to prevent mis-operation with each other. (2) The operator should do the job in a position ready to press the EMERGENCY stop pushbutton switch at any time. Also, he should perform the job in an area from which he can evacuate immediately at the time of an emergency after confirming the robot working range and shields nearby. (3) The supervisor should keep watch on the job at a position where he can see the entire robot system and operate the EMERGENCY stop pushbutton switch at the time of an emergency. Also, he should keep anyone from entering the dangerous area. Unless the operator or other person follows the instructions of the supervisor, an accident will be caused. • If an abnormality has generated or the POWER LED lamp on the control panel remains off after the main power switch of the equipment was turned on, turn off the main power immediately and confirm the wiring. Otherwise, you will be struck by an electric shock or a fire will break out. • Unless the robot operates toward a designated direction at manual guide, turn off the servo power. Otherwise, the robot will be damaged or you will be involved in an accident. When this happens, call us at the after-sale service agent. • Pushbutton operations on the control panel and teach pendant should be confirmed visually. Otherwise, you will be involved in an accident due to mis-operation. • After the power is turned on, be sure to reset a program to start an automatic operation. If the program is executed continuously, the robot will interfere with the peripheral equipment, resulting in damages or accidents.

CAUTION	
 Mandatory	• Before operating the equipment, perform the following inspection. (1) Make sure that there are no problems on the external appearance of the robot, controller, peripheral equipment and cables. (2) Make sure that no obstacle stands in or near the working range of the robot and peripheral equipment. (3) Make sure that the emergency stop and other safety devices operate properly. (4) Make sure that no abnormal noise or vibration is involved in the robot operation. If the above prior inspection is skipped, the equipment will be damaged or you will be involved in an accident.
 Caution	• The speed of test operations is initially set at 25 % of the maximum robot speed. • The speed of automatic operation is initially set at 100 % of the maximum robot speed.

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Chapter 1 Overview

1.1 What is sensor-less compliance control?

"Sensor-less compliance control" is technology which detects external force or moment exerted on the robot hand, and flexibly controls the operations of the respective joint actuators so that the robot hand will work with the required force under the flexible control. An external sensor such as a "6-axis force torque sensor" is generally used to detect external force exerted on the robot hand. However, the sensor-less compliance control is the function that performs compliance control, estimating the external force to be imposed on the robot hand on the basis of each axis torque value without using any 6-axis force torque sensor.

1.2 Use of sensor-less compliance control

Parts are assembled one on top of the other in assembly work, during which play between the parts accumulates, causing them to get out of position in many cases. As a result, a part or the robot hand causes interference during the assembly process, and an error such as "008-451 to 008-458: Motor locked" or "008-491 to 008-498: Electronic thermal detection" occurs, by which the robot is stopped frequently.
The position error generated at the arm end can be eliminated by using the sensor-less compliance control function that flexibly controls the robot hand to softly touch the parts.

Interlocking:
The robot arm makes a slow approach until it comes in contact with the part, and the hand presses the part with a constant force when touching it.

Inserting:
The robot arm is controlled flexibly in the direction in which it should softly touch the part, and it presses the part with a constant force in the insertion direction.

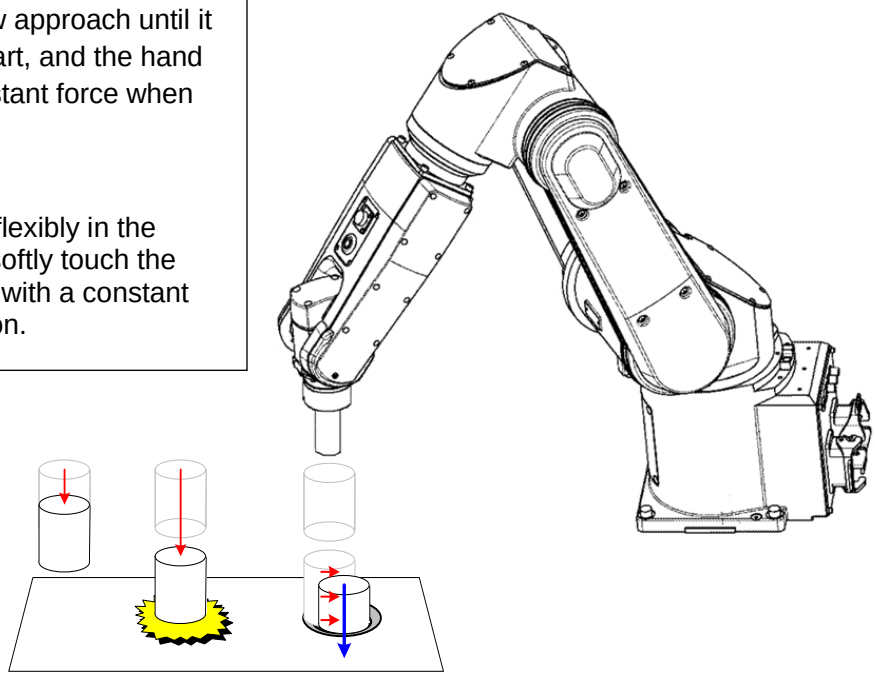


Figure 1-1 Example of work with the sensor-less compliance control function

Chapter 2 Parameters

2.1 Force detection limit

```
[U27] FORCE DETECTION LIMIT [N][Nm]
{+ Direction}
=  0.0   0.0   0.0   0.0   0.0   0.0   0.0   0.0
{- Direction}
=  0.0   0.0   0.0   0.0   0.0   0.0   0.0   0.0
```

Setting : = (x-axis) (y-axis) (z-axis) (a-axis) (b-axis) (c-axis) (u-axis) (v-axis)

Data format : Real number type

Data unit : N or N·m

Data range :

Number of data items : 8

Description : This parameter sets the limit for external force to be detected under compliance control. The detection of the external force limit can be invalidated by setting "0.0" to all the values in both the positive (+) and negative (-) directions. If the detected external force exceeds any of the parameter setting values, a level-8 alarm occurs. Note that value setting for any of the axes A, B, C, U, and V turns out to be invalid under the existing conditions.

Note: Put a negative sign in front of the limit value to be set in the negative direction.

2.2 Position shift limit

```
[U28] POSITION SHIFT LIMIT [mm][deg]
{+ Direction}
=  0.0   0.0   0.0   0.0   0.0   0.0   0.0   0.0
{- Direction}
=  0.0   0.0   0.0   0.0   0.0   0.0   0.0   0.0
```

Setting : = (x-axis) (y-axis) (z-axis) (a-axis) (b-axis) (c-axis) (u-axis) (v-axis)

Data format : Real number type

Data unit : mm or degree

Data range :

Number of data items : 8

Description : This parameter sets the maximum value for each axis in the positive direction {+Direction}, and the minimum value in the negative direction {-Direction}, which are to be used when the position is corrected through compliance control. The detection of external force limit can be invalidated by setting "0.0" to all the values in both the positive (+) and negative (-) directions. The parameter can select whether the limit is valid or invalid for each axis. If the detected value exceeds the setting value, a level-8 alarm occurs. Note that value setting for any of the axes A, B, C, U, and V turns out to be invalid under the existing conditions.

Note: Put a negative sign in front of the limit value to be set in the negative direction.

Chapter 3 Compliance Control Library

This chapter explains how to use the library program functions necessary for performing sensor-less compliance control.

3.1 Instruction word format

Subprograms in the compliance control library are explained in the following format in this section.

(1) Function
Briefly explains the functions of subprograms in the compliance control library.

(2) Format
Explains the description format of subprograms in the compliance control library.
The symbols used in the explanation have the following meanings:

[ ]	:	Indicates that items enclosed in brackets can be omitted. Specify them as needed.
< >	:	Indicates the details of the data to be written.
{ }	:	Indicates that only 1 of items enclosed in braces can be selected.
. . .	:	Indicates that more than 1 data item can be specified.

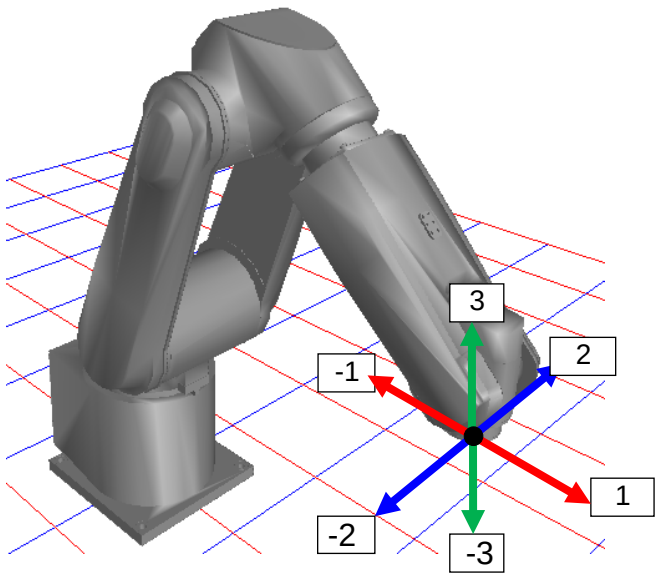
The above symbols are added for the sake of of explanation. Do not write them in actual programs. If use of the above symbols is necessary in a particular case, there would be an explanation accordingly.

(3) Sample statement
Indicates sample program description using the compliance control library.

(4) Comment / Note
Explains comments on the compliance control library, and notes or restrictions on the use of instruction words.

CMPTOUCH

Function	Allows the robot arm to press parts through sensor-less compliance control.
Format	CMPTOUCH(<pressing-direction>,<compliance-gain>,<target-force>, <pressing-time>)
Sample Statement	CMPTOUYCH (3, 2, 3, 60.0)
Comment • Note	When CMPTOUCH is executed, the robot or tool end moves until force is detected in a given direction. After force detection, the robot hand presses the part for a preset period of time. 4 values must be set for executing this command. For <pressing-direction>, determine the pressing direction with an axis direction in the robot tool coordinate system. Specify the direction with an integer of 1 to 3 or -1 to -3. Figure 3-1 shows the pressing directions corresponding to the respective values.



Value	Axis direction in the tool coordinate system
1	X coordinate direction
-1	-X coordinate direction
2	Y coordinate direction
-2	-Y coordinate direction
3	Z coordinate direction
-3	-Z coordinate direction

Figure 3-1 Pressing directions corresponding to the setting values

For <compliance-gain>, specify the sensitivity for external force with an integer 0 to 5.
As the value is greater, the robot corrects the position with higher sensitivity to external force.

For <target-force>, determine the force of pressing the part. The strength of pressing force can be set at 5 levels; specify the force with an integer 1 to 5. As with the case of compliance gain, the robot arm approaches the part faster as the specified value increases.
If the recommended operating conditions described later are satisfied, the target force is as shown in Table 3-1, corresponding to the respective setting values.

Table 3-1 Relationships between setting values and target force

Setting value	1	2	3	4	5
Target force	About 30 N	About 40 N	About 50 N	About 60 N	About 70 N

Note: The table shows the rough standards of the pressing force, but not the guaranteed values.

For <pressing-time>, specify the time (sec) for pressing the part after the detection of external force. The time must be specified with a value of up to 200.0. (The value 0.0 cannot be specified.) If the robot arm moves before the preset pressing time has passed, the elapsed pressing time is reset, and when external force is detected next time, control waits until the pressing time has passed. If any of the setting values is outside the range or if the robot hand does not touch the part after moving 200.0 mm or more during the execution of the CMPTOUCH command, control stops the command execution, and moves to the next command.
The CMPTOUCH command is written in the compliance control library (COMPCTRL.LIB). To use the command, call COMPCTRL.LIB through LOADLIB.

Caution	Be sure to specify the "tool coordinate system" and "load condition" of the hand to be used when programming compliance control. If these are not specified correctly, the robot may make an unexpected motion during compliance control. • Specify correct tool offset in the TOOL command. • Specify the hand (+ part) load condition in the PAYLOAD command.

Sample program

```
GROBAL
   LOADLIB COMPCTRL. LIB
END
PROGRAM COMP
   TPPL = T1
   PAYLOAD = {2,  0,  0.  0,  0.  0,  -100.  0,  0.  0,  0.  0,  0.  0}
   MOVE A1
   MOVE A2
   CMPTOUCH (-3,  2,  3,  60.  0)
   MOVE A3
END
```

Explanation of CMPTOUCH library internal processing

```
PROGRAM CMPTOUCH1( C_AXISDI, C_CGAINNUM, C_TGETFCNUM, C_TTIME)

  DELAY 0.001
  WAIT MOTION >= 100
  '----- Compliance Parameter Setting -----
  C_OMG = 5.000                    ' Spring component
  C_ZET = 5.500                    ' Damper component
  C_GKI = 0.5                      ' Compliance-control integral gain
  '----- Compliance Control Threshold Setting -----
  C_MAXTTIME = 200.000             ' Maximum pressing time   =  200 sec
  C_MAXDIFF  = 200.000             ' Maximum travel distance =  200 mm
  C_STPSPD   =   0.040             ' Stop determination speed =  0.004 mm/cycle
  C_SIGN     =  -1.000
  C_ERROR_FLG=0
  '----- Parameter Error Check (C_ERROR_FLG==2) -----
  'C_AXISDI : Pressing direction (1:X, -1:-X, 2:Y, -2:-Y, 3:Z, -3:-Z)
  IF C_AXISDI < -3 OR C_AXISDI > 3 THEN C_ERROR_FLG = 2
  IF C_AXISDI > -1 AND C_AXISDI < 1 THEN C_ERROR_FLG = 2
  'C_CGAINNUM : Gain number (1-5)
  IF C_CGAINNUM < 0 OR C_CGAINNUM > 5 THEN C_ERROR_FLG = 2
  'C_TGETFCNUM : Pressing force (1-5)
  IF C_TGETFCNUM < 1 OR C_TGETFCNUM > 5 THEN C_ERROR_FLG = 2
  'C_TTIME : Pressing time > C_MAXTTIME
  IF C_TTIME <= 0 OR C_TTIME > C_MAXTTIME THEN C_ERROR_FLG = 2
  '----- FHT Error Check (C_ERROR_FLG==1) -----
  IF FHT_NO <> 324 THEN C_ERROR_FLG = 1     ' The FHT number must be 324.
  IF FHT_VER < 6.00 THEN C_ERROR_FLG = 1    ' The FHT version must be 6.00 or later.
  '----- Error Display on the TP -----
  IF C_ERROR_FLG == 1 THEN PRINT TP,"ROBOT PARAMETER ERROR",CR
  IF C_ERROR_FLG == 2 THEN PRINT TP,"COMP  PARAMETER ERROR",CR
  IF C_ERROR_FLG >= 1 THEN GOTO C_JP0       ' Occurrence of error -> Jump to the end
  '----- Pressing Force Direction Setting ----
  C_AXIS = C_AXISDI
  IF C_AXISDI > 0 THEN GOTO C_JP3
  C_AXIS = C_SIGN * C_AXISDI                ' Sign conversion
  C_SIGN=1.0
C_JP3:
  '----- Target Pressing Force and Compliance Gain Settings -----
  C_DFC = C_SIGN*(20+10*C_TGETFCNUM)     ' Set the target pressing force to "C_TGETFCNUM."
  C_GKG = C_CGAINNUM*0.06                ' Set the compliance gain to "C_CGAINNUM."
  '----- Compliance Parameter Setting -----
  GOTO(C_AXIS) C_XAXIS, C_YAXIS, C_ZAXIS       ⇒ To the next page
```

```
C_XAXIS:
  COMP = 0 ' The compliance control coordinate system is set with the tool coordinate system. (Change not allowed)
  DFORCE = {  C_DFC,   0.000,   0.000,   0.000,   0.000,   0.000,   0.000,   0.000 }
  CMPGKG = {  C_GKG,   0.000,   0.000,   0.000,   0.000,   0.000,   0.000,   0.000 }
  CMPOMG = {  C_OMG,   C_OMG,   C_OMG,   0.000,   0.000,   0.000,   0.000,   0.000 }
  CMPZET = {  C_ZET,   C_ZET,   C_ZET,   0.000,   0.000,   0.000,   0.000,   0.000 }
  CMPGKI = {  C_GKI,   C_GKI,   C_GKI,   0.000,   0.000,   0.000,   0.000,   0.000 }
  GOTO C_JP1
C_YAXIS:
  COMP = 0  ' The compliance control coordinate system is set with the tool coordinate system. (Change not allowed)
  DFORCE = {  0.000,   C_DFC,   0.000,   0.000,   0.000,   0.000,   0.000,   0.000 }
  CMPGKG = {  0.000,   C_GKG,   0.000,   0.000,   0.000,   0.000,   0.000,   0.000 }
  CMPOMG = {  C_OMG,   C_OMG,   C_OMG,   0.000,   0.000,   0.000,   0.000,   0.000 }
  CMPZET = {  C_ZET,   C_ZET,   C_ZET,   0.000,   0.000,   0.000,   0.000,   0.000 }
  CMPGKI = {  C_GKI,   C_GKI,   C_GKI,   0.000,   0.000,   0.000,   0.000,   0.000 }
  GOTO C_JP1
C_ZAXIS:
  COMP = 0  ' The compliance control coordinate system is set with the tool coordinate system. (Change not allowed)
  DFORCE = {  0.000,   0.000,   C_DFC,   0.000,   0.000,   0.000,   0.000,   0.000 }
  CMPGKG = {  0.000,   0.000,   C_GKG,   0.000,   0.000,   0.000,   0.000,   0.000 }
  CMPOMG = {  C_OMG,   C_OMG,   C_OMG,   0.000,   0.000,   0.000,   0.000,   0.000 }
  CMPZET = {  C_ZET,   C_ZET,   C_ZET,   0.000,   0.000,   0.000,   0.000,   0.000 }
  CMPGKI = {  C_GKI,   C_GKI,   C_GKI,   0.000,   0.000,   0.000,   0.000,   0.000 }
C_JP1:
  CMPTCHSP = PSNCMD     ' The start position is acquired.
  C_DIFF = 0
  C_DIFFOLD = 0
  ENABLE COMP           ' Compliance control start
  DELAY 0.001
  TIMER = C_TTIME       ' Set the pressing time to C_TTIME.
C_JP2: '----- Completion of Compliance Parameter Setting -----
  C_TIME = TIMER - 0.050                        '0.05 (sec) waiting
  IF (C_TIME < 0.000) THEN C_TIME = 0.000
  WAIT TIMER <= C_TIME
  C_DIFFOLD = C_DIFF                            ' The travel distance after 0.05 sec is stored.
  C_CMPTCHEP = PSNCMD                           ' The current position is acquired.
  C_DIFFSQ = (CMPTCHSP.X-C_CMPTCHEP.X)*(CMPTCHSP.X-C_CMPTCHEP.X)
    +(CMPTCHSP.Y-C_CMPTCHEP.Y)*(CMPTCHSP.Y-C_CMPTCHEP.Y)
+(CMPTCHSP.Z-C_CMPTCHEP.Z)*(CMPTCHSP.Z-C_CMPTCHEP.Z)
  C_DIFF = SQRT(C_DIFFSQ)           ' The travel distance "C_DIFF" from the start position is calculated.
  IF ( C_DIFF >= C_MAXDIFF) THEN GOTO C_JP0 ' The travel distance from the start position is checked.
  IF ( TIMER <= 0.000) THEN GOTO C_JP0    ' The pressing time is checked.
  IF ( ABS(C_DIFF - C_DIFFOLD) > C_STPSPD) THEN TIMER = C_TTIME    ' The motion speed is checked.
  GOTO C_JP2
C_JP0: '----- End of Compliance Control -----
   DISABLE COMP
  DELAY 0.001
END
```

Chapter 4 Notes

This chapter explains notes on performing sensor-less compliance control.

4.1 Recommended operating conditions

It is recommended that the robot operate under the following conditions when the hand is to press parts through sensor-less compliance control. Robot operation under the following conditions enables the hand to press parts in a more stable manner.

4.1.1 When pressing parts in a direction vertical to the installation surface:

If the robot hand is to press parts in the direction vertical to the robot installation surface (i.e., in the negative Z direction in the world coordinate system) as shown in Figure 4-1, the angle of the axis 2 is recommended to be 40° or more.

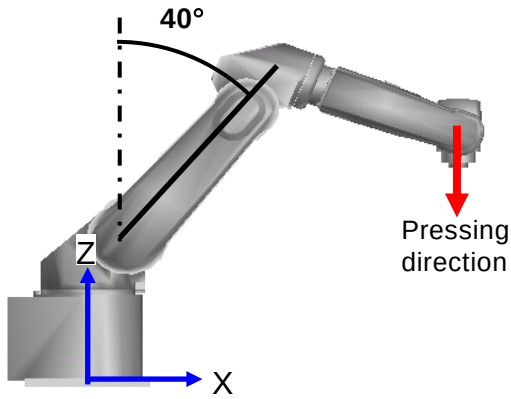


Figure 4-1 Pressing in the vertical direction

4.1.2 When pressing parts in a direction horizontal to the installation surface:

If the robot hand is to press parts in the direction horizontal to the robot installation surface as shown in Figure 4-2, the angle of the axis 3 is recommended to be 40° or more.

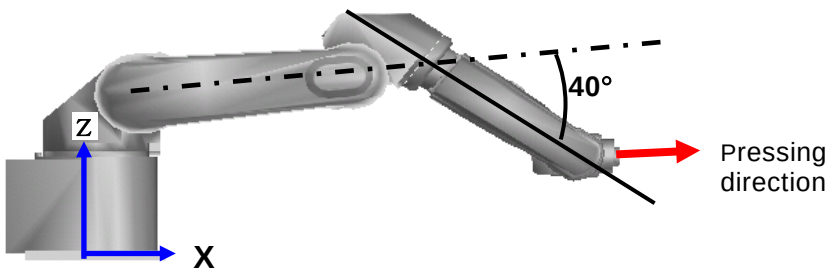


Figure 4-2 Pressing in the horizontal direction

4.1.3 When pressing parts in a direction horizontal to the robot installation surface:

As the pressing direction further departs from the central axis direction of the arm, the robot hand presses parts more unstably. Pressing in the same direction as the central axis direction is recommended.

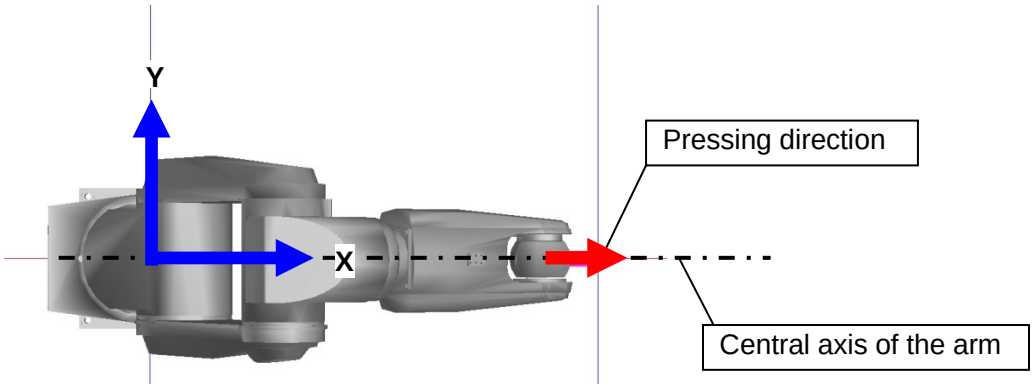


Figure 4-3 Pressing in the same direction as the central axis direction of the arm (stable pressing)

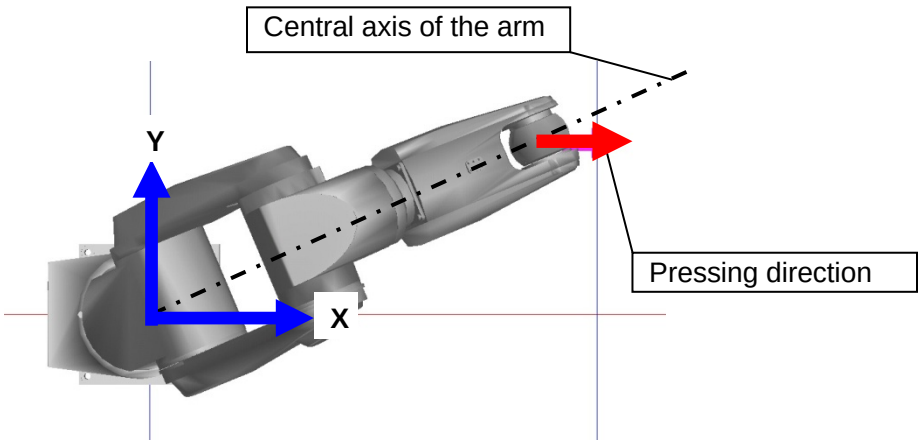


Figure 4-4 Pressing in a direction different from the central axis direction of the arm (unstable pressing)

4.1.4 Motion in the vicinity of singularities

If a robot posture is in the vicinity of singularities, the robot hand might fail to press parts properly, in which case the use of compliance control should be avoided. There are the following 2 types of singularities:

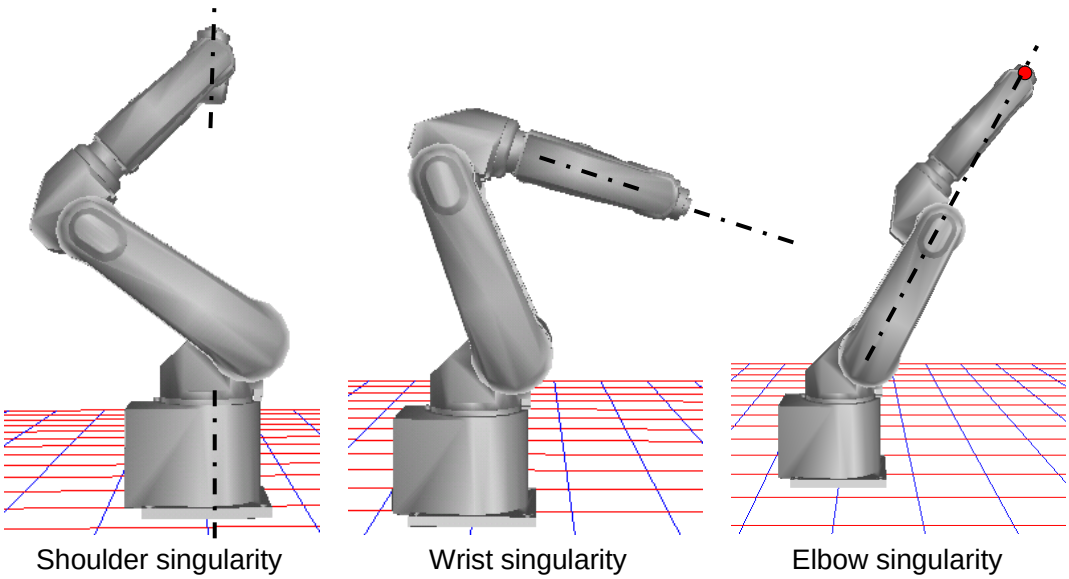


Figure 4-4 Singularities of TV-series robots

Shoulder singularity:

Means a posture in which the rotation centers of the axes 1 and 6 are aligned in tandem.

Wrist singularity:

Means a posture in which the rotation centers of the axes 4 and 6 are aligned in tandem (that is, the axis 5 is at 0°).

Elbow singularity:

Means a posture in which the elbow is fully stretched (that is, the central axis of arm 1 is aligned with the center of axis 5).

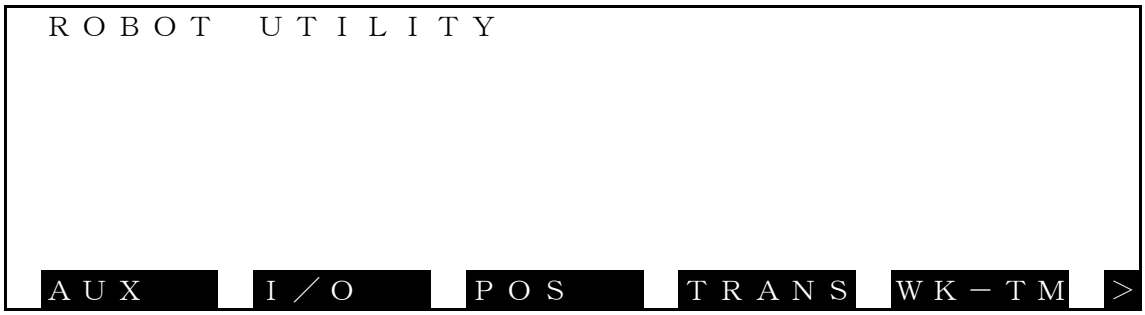
Chapter 5 Utilities

The utility mode enables the display of external input and output signals, the display of the current position, the selection of workpiece/tool coordinates, and the like. In addition, it can be used in any of the manual, internal automatic, and external automatic modes.
Pressing ESC terminates the utility mode and returns to the original screen.

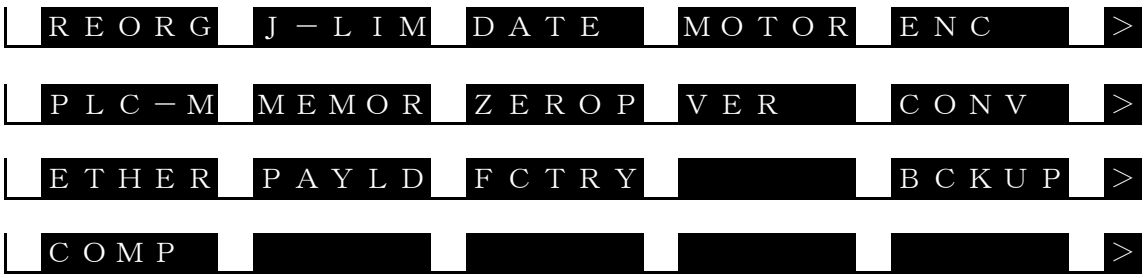
5.1 Utility mode selection

(1) Function
Switches to the utility mode.

(2) Operation procedure
Step 1: Utility mode selection
Press UTILITY. The display screen as below is shown.



The menu items switches as below each time you press NEXT.



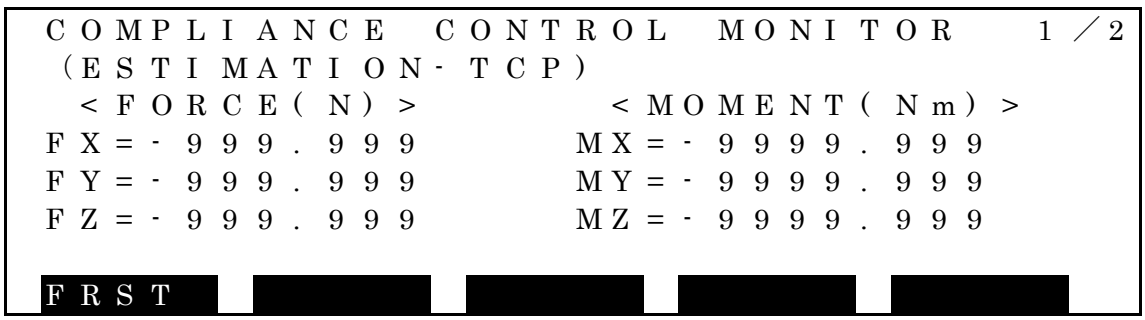
(3) Note
The utility mode can be used in any of the manual, internal automatic, and external automatic modes.

5.2 Compliance control status screen [COMP]

(1) Function
Allows users to check the states of compliance control.

(2) Operation procedure

(a) Step 1: File mode selection
Hold down NEXT until [COMP] is displayed, and press [COMP].
The hand input/output signal status display screen as shown below is displayed.



See step 2 for the display contents.

Shown below are the valid keys and the key functions.

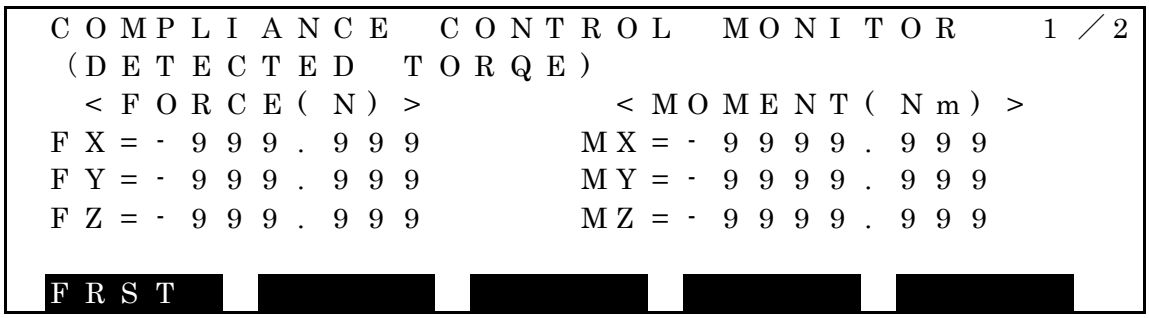
Key	Function
ALT + NEXT or ↓	Switches to the next page. Switches to the first page if the current page is the last page.
NEXT or ↑	Switches to the previous page. Switches to the last page if the current page is the first page.
ESC	Returns to the Utility screen.

Shown below is the function of a command in the command menu.

Command	Function
[FRST]	Resets the detected value to zero (0) at the execution of FRST.

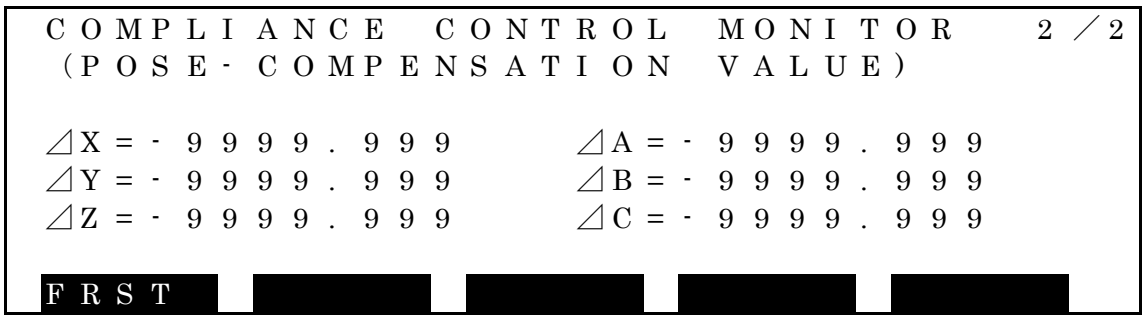
(b) Step 2: Screen switching
The display items are changed by pressing ALT + NEXT or ↓ (alternatively NEXT or ↑).
The following shows the display items on each page.

Page 1 (Values of estimated external force and moment (tool coordinate system))



- The page displays estimated external force and moment exerted on the tool end.
- The values of estimated external force and moment are displayed in units of N for the X-axis, Y-axis, and Z-axis directions; they are displayed in units of N·m for the A-axis, B-axis, and C-axis directions.

Page 2 (compliance compensation value)



- The page displays the amounts of position shift under compliance control.
- The values of compliance compensation are displayed in millimeters for the X-axis, Y-axis, and Z-axis directions; they are displayed in degrees for the A-axis, B-axis, and C-axis directions.

Chapter 6 Alarms

Error code	8－289 Fx Detection Limit 8－290 Fy Detection Limit 8－291 Fz Detection Limit 8－292 Mx Detection Limit 8－293 My Detection Limit 8－294 Mz Detection Limit
Name	**External force detection error in the X-axis, Y-axis, Z-axis, A-axis, B-axis, or C-axis direction**
Description	Any of the detected external force exceeded the value set to the parameter [U27].
Cause	• The value of the user parameter [U27] FORCE DETECTION LIMIT is too small. • A wrong sign "±" is set to the setting value. • The target force or compliance gain is too large during compliance control.
Corrective measures	• Increase the parameter setting value ([U27]) of the axis with which the alarm was detected. • Check the sign. No sign is necessary for values in the positive axis directions. Assign a negative sign for values in the negative axis directions. • If the robot oscillates during operation under compliance control, set a smaller value to the target force or compliance gain.
Remarks	• This version X8GCAV-06A does not perform compliance control in the rotary axis direction, so the alarms 8-292, 8-293, and 8-294 do not occur. • To disable the detection of the above alarms, set 0.0 to all the values of the user parameter [U27] FORCE DETECTION LIMIT.

Error code	8－297 Dx COMP Control Limit 8－298 Dy COMP Control Limit 8－299 Dz COMP Control Limit 8－300 Da COMP Control Limit 8－301 Db COMP Control Limit 8－302 Dc COMP Control Limit
Name	**Position shift amount too large for the X-axis, Y-axis, Z-axis, A-axis, B-axis, or C-axis direction**
Description	Any of the amounts of position shift through compliance control exceeded the value set to the parameter [U28].
Cause	• The value of the user parameter [U28] POSITION SHIFT LIMIT is too small. • A wrong sign "±" is set to the setting value. • The parameter cannot yet be disabled. • The robot moved away from the part the hand intended to press during compliance control.
Corrective measures	• Increase the parameter setting value ([U28]) of the axis with which the alarm was detected. • Check the sign. No sign is necessary for values in the positive axis directions. Assign a negative sign for values in the negative axis directions. Confirm that the values for both the positive and negative directions are 0.000. • Check if the compliance control start position is correct and if the part to be pressed is set correctly. • If the robot moves in an unexpected direction during compliance control, check the setting of the pressing direction. If the robot oscillates when the hand is touching the part, set a smaller value to compliance gain.
Remarks	• This version does not perform compliance control in the rotary axis direction, so the alarms 8-300, 8-301, and 8-302 are not detected. • To disable the detection of the above alarms, set 0.0 to all the values of the user parameter [U28] POSITION SHIFT LIMIT.

Error code	2 – 0 3 8 COMP command error
Name	Compliance control command execution error
Description	Compliance control command cannot be executed
Cause	• Zero (tool coordinate system) is not set at the selection of a coordinate system for compliance control. • A command other than DELAY is used as the command to start compliance control.
Corrective measures	• Set 0 (tool coordinate system) at the selection of a coordinate system for compliance control. • Use DELAY as the command to start compliance control.
Remarks	